

# evidence

## **BOB-066 Evidence — Cross-Layer BOBA\_UPSTREAM\_PROXY Audit**

**Revision:** 1 **Last modified:** 2026-08-15T12:15:00Z **Item:** BOB-066 — Lava P3: BOBA\_UPSTREAM\_PROXY in download-proxy + qBitTorrent-go + Jackett + compose env-forward **Status:** In progress (L1/L2/L4-partial WIRED; L3 residual gap) **Verifier:** challenges/scripts/upstream\_proxy\_wired\_challenge.sh **Run id:** 20260815T115831Z

### **§11.4.108 Runtime signature — three-mode verifier evidence**

Anti-bluff coverage per §11.4.115 polarity switch + §11.4.107(10) self-validation. Each mode's exit code is captured; every mode required to exit 0 for closure eligibility.

#### **Mode 1 — --self-validate (analyzer-honesty against a synthetic golden-BAD tree)**

The verifier runs its OWN detector chain against a temp tree where every wire is DELIBERATELY ABSENT (stub main.py without apply\_proxy\_env, stub search.py with zero trust\_env, stub httpx/proxy.go with no Configure/NewTransport/Proxy, empty docker-compose.yml). PASS = every detector correctly returned FAIL on the bad tree → the analyzer is not a bluff.

```
=== self-validate: run detectors against a synthetic golden-BAD tree ===
```

```
PASS: self-validate – every detector correctly FAILED on the golden-BAD tree
```

**Exit: 0** — analyzer honesty confirmed.

## **Mode 2 — GREEN (RED\_MODE=0, default) — real tree**

=== BOB-066 four-layer cross-layer verifier ===

RED\_MODE=0 (0=GREEN guard, 1=reproduce pre-fix defect state)

--- L1: download-proxy (Python) ---

[PASS] L1 config/proxy.py exports apply\_proxy\_env +  
aiohttp\_session\_kwarg

[PASS] L1 apply\_proxy\_env() invoked at boot in download-proxy/  
src/main.py

[PASS] L1 aiohttp ClientSession sites carry trust\_env (9 uses of  
\_tracker\_session\_kwarg)

--- L2: qBitTorrent-go ---

[PASS] L2 internal/httpx/proxy.go exports Configure +  
NewTransport + Proxy honoring BOBA\_UPSTREAM\_PROXY

[PASS] L2 httpx.Configure(cfg.UpstreamProxy) called in cmd/  
qbittorrent-proxy/main.go

[PASS] L2 httpx.NewTransport() used in internal/api (torrent-  
download client)

--- L3: Jackett ---

[SKIP] L3 Jackett proxy honor: reason=extension\_absent (neither  
compose env-forward nor ServerConfig.ProxyUrl wired) – tracked as  
BOB-066 residual

--- L4: docker-compose.yml env-forward ---

[PASS] L4 docker-compose.yml service 'download-proxy' env-  
forwards proxy vars

[PASS] L4 docker-compose.yml service 'qbittorrent-proxy-go' env-  
forwards proxy vars

[PASS] L4 docker-compose.yml service 'qbittorrent' env-forwards  
proxy vars

[SKIP] L4 'jackett' env-forward absent (see L3 SKIP; tracked)

[SKIP] L4 'boba-jackett' env-forward absent (not tracker-bound:  
talks to Jackett loopback + local DB only)

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Summary: PASS=9 FAIL=0 SKIP=3

Notes:

- L3 residual: Jackett has neither env-forward nor  
ServerConfig.ProxyUrl wired – see BOB-066 description

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PASS: upstream\_proxy\_wired\_challenge – every wired layer verified;  
residual gaps honestly SKIP'd

**Exit: 0** — 9 PASS / 0 FAIL / 3 honest SKIP (§11.4.3 / §11.4.69  
extension\_absent).

## Mode 3 — RED (RED\_MODE=1) — §11.4.115 polarity flip

Under RED each presence-detector's polarity flips: PRESENT code becomes FAIL (because the pre-fix defect state RED simulates would have it ABSENT). On a tree where the fix has landed, EXPECT many FAILs — that IS the signal the polarity switch + detectors work end-to-end (they would have caught the defect if the code were un-wired). `fail == 0` under RED would mean detectors ignore polarity OR the tree was reverted — both bluffs surfaced here.

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Summary: PASS=0 FAIL=9 SKIP=3

=====

RED-MODE VERIFY: 9 presence-detectors correctly negated under RED  
– polarity switch works, RED→GREEN flip proven

**Exit: 0** — 9 detectors correctly negated → RED→GREEN flip proven.

## §11.4.108 four-layer audit — per-layer runtime signatures

| Layer | Sub-requirement                                                                                 | Runtime signature                                                                                                                                                                                                                                                                                                                                                                | Status                     |
|-------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| L1    | Python download-proxy honors BOBA_UPSTREAM_PROXY (urllib subprocess + aiohttp)                  | download-proxy/src/main.py:97 calls <code>apply_proxy_env()</code> ; 9 <code>aiohttp.ClientSession</code> sites in <code>merge_service/search.py</code> carry <code>_tracker_session_kwargs()=trust_env</code> ; <code>config/proxy.py</code> exports the API + <code>UPSTREAM_PROXY_ENV="BOBA_UPSTREAM_PROXY"</code> sentinel                                                   | WIRED                      |
| L2    | qBitTorrent-go <code>http.Transport.Proxy</code> honors BOBA_UPSTREAM_PROXY (or standard chain) | <code>internal/httpx/proxy.go</code> exports <code>Configure/NewTransport/Proxy</code> ; <code>cmd/qbittorrent-proxy/main.go:29</code> calls <code>httpx.Configure(cfg.UpstreamProxy)</code> at boot; <code>internal/api/download.go:65</code> installs <code>httpx.NewTransport()</code> on the torrent-download client; <code>socks5://</code> supported natively (remote DNS) | WIRED                      |
| L3    | Jackett — env-forward OR <code>ServerConfig.ProxyUrl</code>                                     | Neither path present. <code>jackett</code> container has no proxy env-forward; no Go code touches <code>ProxyUrl/ProxyType/ServerConfig</code> in <code>qBitTorrent-go/internal/</code> .                                                                                                                                                                                        | RESIDU GAP                 |
| L4    | <code>docker-compose.yml</code> env-forward for every tracker-touching service                  | <code>download-proxy</code> , <code>qbittorrent-proxy-go</code> , <code>qbittorrent</code> all env-forward BOBA_UPSTREAM_PROXY + HTTP_PROXY + HTTPS_PROXY + NO_PROXY (with default                                                                                                                                                                                               | 3/5 WIRED (Jackett gap = L |

| Layer | Sub-requirement | Runtime signature                                                                             | Status |
|-------|-----------------|-----------------------------------------------------------------------------------------------|--------|
|       |                 | bypass hosts). jackett + boba-jackett do NOT (boba-jackett is loopback-only so not required). |        |

## Recommended follow-up for L3

File a scoped ticket (e.g. BOB-066b) implementing ONE of: - **(a)** Add proxy env-forward to the jackett service block in docker-compose.yml (mirror the block at lines 15-20 of the qbittorrent service). Simplest fix. - **(b)** Add qBitTorrent-go/internal/jackettconfig/proxy.go that on Jackett-config load reads BOBA\_UPSTREAM\_PROXY and PATCHes Jackett's ServerConfig.json ProxyType/ProxyUrl fields (Jackett will honor them for its own indexer HTTP egress). Requires a boba-jackett boot-time step + safe file-write discipline.

## Status decision (§11.4.6 / §11.4.108 / §11.4.226)

BOB-066 is NOT closed as Completed because L3 remains un-wired: closing with 3/4 sub-requirements met would be a §11.4.226 evidence-class-at-closure violation (the reported defect layer includes L3, and a source-green closure of an un-wired layer is exactly the pattern the anchor forbids). Bumped Queued → In progress; item\_history entry #65 records the audit with this evidence path; description carries the AUDIT NOTE 2026-08-15 block naming L1/L2/L4 wired + L3 residual + follow-up recommendation.